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PROTECTIVE COMPOSITE FABRICS AND METHODS OF MANUFACTURE AND USE

Abstract

A composite fabric includes a film, a first fibrous layer, a fabric layer, and a second fibrous layer and one or more retaining members. The film has a first side and a second side. The first fibrous layer has a first side connected to the second side of the film and a second side. The fabric layer has a first side connected to the second side of the first fibrous layer and a second side. The second fibrous layer has a first side connected to the second side of the fabric layer and a second side. The film can be a non-heat shrinking film. The retaining members may be configured to contact a surface on which the composite fabric is placed and resist movement of the composite fabric relative to the surface where the retaining members include a corrosion inhibitor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/406,846, filed Jan. 8, 2024, entitled “PROTECTIVE COMPOSITE FABRICS AND METHODS OF MANUFACTURE AND USE,” which is a continuation of U.S. patent application Ser. No. 18/333,022, filed Jun. 12, 2023, entitled “PROTECTIVE COMPOSITE FABRICS AND METHODS OF MANUFACTURE AND USE,” which is a continuation of U.S. patent application Ser. No. 17/324,450, filed May 19, 2021, entitled “PROTECTIVE COMPOSITE FABRICS AND METHODS OF MANUFACTURE AND USE,” which claims priority to U.S. Provisional Application No. 63/028,814, filed on May 22, 2020, entitled “PROTECTIVE COMPOSITE FABRICS AND METHODS OF MANUFACTURE AND USE,” the entire disclosures of which are incorporate herein by reference.

BACKGROUND

[0002] The present invention relates to protective composite fabrics and methods of manufacturing and using protective composite fabrics.

[0003] Protective fabrics and other constructs can be used to prevent corrosion, scratches and other damage to surfaces and objects that can result from contact with other objects, rain, wind, sunlight, the elements and other sources of harm. Examples of such fabrics and constructs are described in U.S. Pat. Nos. 6,696,120, 7,074,288, 8,828,487 and 8,883,284.

SUMMARY

[0004] One embodiment as shown and described herein may include a composite fabric that includes a film having a first side and a second side, a first fibrous layer having a first side facing the second side of the film and a second side, a fabric layer having a first side facing the second side of the first fibrous layer and a second side, a second fibrous layer having a first side facing the second side of the fabric layer and a second side, and a plurality of retaining members extending from the second side of the second fibrous layer and configured to contact a surface on which the composite fabric is placed and to space at least a portion of the second side of the second fibrous layer from the surface, wherein the retaining members include a first vapor corrosion inhibitor within the plurality of retaining members configured to vaporize and collect between the surface and the portion of the second side of the second fibrous layer spaced from the surface wherein the plurality of retaining members and the second side of the second fibrous layer are configured to cooperate with the surface to form at least one pocket configured to retain the vaporized first vapor corrosion inhibitor therein.

[0005] Another embodiment as shown and described herein may further or alternatively include a composite fabric including a first film having a first side and a second side, a first fibrous layer having a first side facing the second side of the film and a second side, a fabric layer having a first side facing the second side of the first fibrous layer and a second side, a second fibrous layer

having a first side facing the second side of the fabric layer and a second side, a second film having a second side facing the second side of the second fibrous layer and a second side, and a plurality of retaining members connected to and extending from the second side of the second film and configured to contact a surface on which the composite fabric is placed and space a portion of the second surface of the second film from the surface, wherein the retaining members include a corrosion inhibitor configured to collect between the second film and the surface, and wherein the plurality of retaining members form at least one pocket between one or more of the plurality of retaining members, the second side of the second film, and the surface, and wherein the first corrosion inhibitor vaporizes and is retained within one or more of the pockets.

[0006] Yet another embodiment as shown and described herein may further or alternatively include a composite fabric that includes a first film having a first side and a second side, a fabric layer having a first side and a second side, at least one fibrous layer, a second film having a first side facing the first film layer and a second side, wherein the fabric layer and the at least one fibrous layer is positioned between the first and second film layers, and a plurality of retaining members configured to contact a surface on which the composite fabric is placed, the retaining members connected to and extending from the second side of the second film, wherein the retaining members include a first vapor corrosion inhibitor, and wherein the plurality of retaining members form at least one pocket between one or more of the plurality of retaining members, the second side of the second film, and the surface, wherein at least a portion of the first vapor corrosion inhibitor collects in one or more of the pockets.

[0007] These and other features of the present invention will be apparent to those of skill in the art from the following detailed description of embodiments of the invention and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a cross sectional view of a protective composite fabric according to one embodiment of the present invention;

[0009] FIG. 2 is a partially exploded view of the protective composite fabric shown in FIG. 1;

[0010] FIG. 3 is a side view of a vehicle with a protective cover constructed from protective composite fabric according to one embodiment of the present invention applied thereto and shown in cross section;

[0011] FIG. 4 is a detail of area 4 in FIG. 3; and

[0012] FIG. 5 is a cross sectional view of a protective composite fabric according to another embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0013] For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the invention as oriented in FIGS. 1, 3 and 4. However, it is to be understood that the invention may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0014] FIG. 1 is a cross-sectional view of a protective composite fabric 10 according to one embodiment of the present invention. In the embodiment shown composite fabric 10 generally includes a film 20, adhesive 30, a first fibrous layer 40, a fabric layer 50, a second fibrous layer 60 and a plurality of retaining members 70.

[0015] Film **20** in the embodiment shown includes a first layer **21**, a second layer **22**, and a third layer **23**. Film **20** is produced from a non-heat shrinking thermoplastic material. In certain embodiments of the invention, first layer **21** is a polyurethane film, and in some embodiments, is a high yield strength, thermoplastic polyurethane film. Examples of materials suitable for film **20** in certain embodiments of the invention include, but are not limited to, ESTANE® 58277 TPU available from Lubrizol Advanced Materials, Inc. of Cleveland, Ohio. In the embodiment shown, second layer **22** of film **20** is a polyurethane film. Second layer **22** can be constructed from the same material as first layer **21** or from a different material. Third layer **23** of film **20** in the embodiment shown is also a polyurethane film. In certain embodiments, third layer **23** is a high yield strength polyurethane film. In certain embodiments, first layer **21** and third layer **23** are produced from the same material. Film **20** may be produced, for example, by coextruding first layer **21**, second layer **22** and third layer **23** together. Film **20** can be produced by extruding cast film and and/or blown film. In one embodiment of the present invention, first layer **21**, second layer **22** and third layer **23** are coextruded to form film **20**.

[0016] In certain embodiments of the invention, second layer **22** of film **20** is thicker than first layer **21** and third layer **23**. In one embodiment, second layer **22** constitutes approximately 60% of the total thickness of film **20** and first layer **21** and third layer **23** each constitute approximately 20% of the total thickness of film **20**. In another embodiment, second layer **22** constitutes approximately 70% of the total thickness of film **20** and first layer **21** and third layer **23** each constitute approximately 15% of the total thickness of film **20**.

[0017] First layer **21** and/or third layer **23** may optionally include a pH modifier **24**. Relatively higher pH levels (i.e., a more alkaline environment) help prevent the growth of mold and mildew and help protect metal surfaces from corrosion. In use, first layer **21** faces outward from the surface to be protected and inhibits mold and mildew growth on the surface of covers or other constructs made from composite fabric **10** and on fabric layer **50**. Third layer **23** inhibits mold and mildew growth on fabric layer **50**. In various embodiments of the present invention, pH modifiers **24** are utilized in an amount effective to produce a pH level between the composite fabric **10** and the object being protected of between about 8 and about 13. In certain embodiments of the invention, pH modifier **24** is selected to produce a pH of at least 8. Suitable pH modifiers **24** include, but are not limited to, sodium acetate, wollastonite, calcium oxide and/or calcium hydroxide. pH modifiers **24** can be added directly to the layers of film **20** or as a pH modifying component in the form of a masterbatch including one or more pH modifiers **24**.

[0018] Film **20**, in the embodiment shown, is an essentially waterproof construct with a high water vapor transmission rate. Film **20** thereby prevents liquid water from migrating through film **20** to the object protected by composite fabric **10**, while at the same time permitting moisture vapor to pass from near the object through composite fabric **10**. In certain embodiments of the invention, the water vapor transmission rate of film **20** is from about 50 g/m.sup.2/day to about 200 g/m.sup.2/day at 7.5 mils thickness. In one embodiment, the water vapor transmission rate of film **20** is at least 50 g/m.sup.2/day. Utilizing thinner films **20** can result in even greater water vapor transmission rates, including water vapor transmission rate of at least 500 g/m.sup.2/day.

[0019] Adhesive **30** joins film **20** to first fibrous layer **40**. Adhesive **30** may be any adhesives sufficient to bond film **20** to first fibrous layer **40** in a manner that prevents delamination and unacceptable degradation of composite fabric **10** in use, particularly in hot, humid and/or wet environments. Examples of acceptable adhesives include, but are not limited to, polyurethane adhesives and hot melt styrene ethylene butadiene styrene (SEBS) adhesives. In one embodiment of the invention, AL1262 adhesive with a vapor corrosion inhibitor ("VCI") is utilized. In another embodiment of the invention, AL34-149-1 adhesive with VCI is utilized. These adhesives can be obtained from Adherent Laboratories, Inc. of St. Paul, Minnesota.

[0020] Various types of VCI's can be utilized in connection with embodiments of the present invention. These VCI's can be amines, nitrites, nitrates, triazoles, and other volatile compounds.

Certain chemistries are more effective at inhibiting corrosion of certain metals. For example, benzotriazole is a more effective VCI for copper, whereas cyclohexylamine carbonate is more effective for use with steel. Non limiting examples of vapor corrosion inhibitors useful in connection with embodiments of the present invention include: primary, secondary and tertiary aliphatic amines; aliphatic diamines; cycloaliphatic and aromatic amines; polymethylamines; long chain ethanolamines; imidazolines; amine salts, for example those of carbonic, carbamic, acetic, benzoic, oleic, nitrous and chromic acids; acetylenic alcohols; alkyl chromates; organic esters of nitrous acid; organic esters of phthalic acid; organic esters of carbonic acid; nitronaphthalene; nitrobenzene; amides; mixtures of nitrites with urea, urotropine, or ethanolamines; naphthols; thiourea derivatives; heterocyclic compounds **65** such as benzotriazole, tolyltriazole, and mercaptobenzothiazole and their respective salts; nitrated or sulfonated petroleum derivatives; and organic acid derivatives.

[0021] First fibrous layer **40** in certain embodiments of the invention is a spun-laced non-woven material. Acceptable materials for first fibrous layer **40** can be natural or synthetic and include organic and inorganic fibers and bioplastic materials. Examples include, but are not limited to, cotton, viscose, rayon, acetate, polyamid, nylon, polyester, acrylic, polyethylene terephthalate (“PET”), polypropylene (“PP”), polyethylene (“PE”), polylactic acid (“PLA”), polyurethane, glass and/or other fibers suitable for producing spun-laced nonwoven material. In certain embodiments of the invention, the staple fibers used to produce first fibrous layer **40** are crimped staple fibers suitable for carding, hydroentangling, and heat drying. Denier of the fibers utilized can vary depending on the production method. In one embodiment of the invention, the fibers have a denier of less than 3. In certain embodiments, the fibers have a denier between 0.5 and 1.5. The basis weight of first fibrous layer **40** can vary based on the intended use of composite fabric **10**. In one embodiment of the invention, the basis weight of first fibrous layer **40** is between 10 g/m² and 100 g/m². In one embodiment of the invention, the basis weight of first fibrous layer **40** is 25 g/m².

[0022] Multiple samples were prepared to test the level of adhesion between films **20** and first fibrous layers **40** according to embodiments of the present invention. The samples were prepared by applying an adhesive **30** to a film **20** and laminating film **20** to a fibrous layer **40**. One set of samples was prepared using AL34-149-1 adhesive (“Embodiment 1”) and another set of samples was prepared using AL1262 adhesive (“Embodiment 2”). Both adhesives included a vapor corrosion inhibitor. The samples were prepared using the following production specifications:

TABLE-US-00001	Air Pressure	Adhesive Applied	Fibrous Hopper	Adhesive Roller	Film Layer	Sample Amount	Line Hose	Exiting Air	Nip Material	Material Width	Adhesive Speed	Head T	Head Spray	T Pressure	Polyurethane	Polyester	20.32 cm	30 g/m ²	22.86 m/min	168.33° C.	4.83 N/cm ²	190.56° C.	51.71 N/cm ²
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[0023] Because film **20** is produced from a non-heat shrinking thermoplastic material, it will not distort when it comes into contact with the hot adhesive.

[0024] The samples of the present invention were submerged in water for 24 hours and their saturated bond strength adhesion was tested by using a tensile testing machine and compared to their dry bond strength adhesion. The same test was performed on a prior art composite fabric product that utilizes a heat shrinkable film. The average results for five samples of the prior art and five samples of each of two embodiments of the present invention are set forth in the table below, in which “MD” indicates adhesion in the machine direction (i.e., along the length of the sample) and “CD” indicates adhesion in the cross direction (i.e., across the width of the sample). The adhesion results are reported in g/cm.

TABLE-US-00002	Prior Art	Embodiment 1	Embodiment 2	MD	CD	MD	CD	MD	CD	Dry Bond
774.80	714.96	1,101.18	1,227.95	1,114.96	1,202.36	Strength	Adhesion	Wet Bond	24.80	15.75
1,220.87	1,427.95	1,198.03	1,188.58	Strength	Adhesion	% Adhesion	−96.81%	−97.82%	10.88%	16.27%
7.48%	−1.15%	Loss	Gain							

[0025] As shown in the table above, the bond strength adhesion of the prior art product degraded considerably after soaking in water. In contrast the bond strength adhesion of Embodiment 1 increased significantly. Furthermore, the bond strength adhesion of Embodiment 2 increased significantly in the machine direction and showed only minimal degradation in the cross direction.

[0026] Fabric layer **50** may be joined to first fibrous layer **40** by a hydroentanglement process, such as those used in known spunlaced production methods. Unlike prior art devices that utilize a scrim, fabric layer **50** is a continuous membrane of material. Fabric layer **50** is, in certain embodiments of the invention, a non-woven fabric material. Examples of suitable fabric materials for fabric layer **50** include, but are not limited to, knitted fabrics and spunbond fabrics made from polyamid, nylon, polypropylene, polyester, polyamide, PLA and/or polyurethane. Fabric layer **50** in certain embodiments of the invention includes vapor corrosion inhibitors **51**. The VCI's discussed above in connection with adhesive **30** are examples of VCI's suitable for incorporation into fabric layer **50**.

[0027] Second fibrous layer **60** may be made from the same materials and in the same manners as discussed above in connection with first fibrous layer **40**, and may have the same physical properties. In certain embodiments of the invention, second fibrous layer **60** is identical to first fibrous layer **40**. In other embodiments of the present invention, second fibrous **60** and first fibrous layer **40** are different. Second fibrous layer **60** may be joined to fabric layer **50** by a hydroentanglement process.

[0028] As noted above, in certain embodiments of the invention, first fibrous layer **40**, fabric layer **50**, and/or second fibrous layer **60** may be constructed in whole or in part from PLA, such as PLA that is produced from corn or another renewable resource. Because PLA is biodegradable, composite fabrics **10** that include PLA are environmentally friendly.

[0029] Retaining members **70**, in the embodiment shown, are connected to and extend from one side of second fibrous layer **60**. Retaining members **70** function to resist movement, including sliding movement, of composite fabric **10** across the surface of the object being protected by cover **10**. Retaining members **70** are produced from a material that will provide sufficient frictional engagement with the surface of the object being protected by composite fabric **10**. In one embodiment of the invention, retaining members **70** are made of polyurethane. In certain embodiments of the invention, retaining members **70** are constructed from formulation FL2030 available from Key Polymer Corporation of Lawrence, Massachusetts. In other embodiments, retaining members **70** are made from epoxy. Retaining members **70** may be produced by utilizing a thermoset or a thermoplastic ink that is printed onto second fibrous layer **60**. The ink used to produce retaining members **70** preferably has a viscosity of 30,000 centipoise or higher at ambient temperature, which assists formation of raised retaining members **70**. Certain samples of embodiments of the present invention were partially produced at Spectra Coating Corp. of Leominster, Massachusetts on equipment that printed and dried retaining members **70** on second fibrous layer **60**. Retaining members **70** may be printed onto second fibrous layer **60** either before or after laminating to film **20**.

[0030] Retaining members **70** may include corrosion inhibitors **71**, which may be VCI's or contact corrosion inhibitors. Contact corrosion inhibitors resist metal corrosion via contact with the metal. In embodiments of the invention that utilize corrosion inhibitors **71**, the ink used to produce retaining members **70** has a pH of about 8 or higher, which results in better corrosion inhibition. The VCI's discussed in connection with vapor corrosion inhibitors **51** above are also suitable for use as corrosion inhibitors **71**. Corrosion inhibitors **71** need not be the same as vapor corrosion inhibitors **51**. If contact corrosion inhibitors are used, they can be located on or near the outer surface of retaining members **70**.

[0031] Composite fabric **10** can be formed in any one of a number of manners. In one embodiment of the invention, first fibrous layer **40**, fabric layer **50**, and second fibrous layer **60** are joined together as, for example, by a hydroentanglement process as described above. Retaining members **70** are then applied to second fibrous layer **60**, such as by printing as described above. Film **20** is

then laminated to first fibrous layer **40** as described above. Alternatively, first fibrous layer **40**, fabric layer **50**, and second fibrous layer **60** are first joined together, followed by laminating film **20** to first fibrous layer **40**. Retaining members **70** are then applied to second fibrous layer **60**. [0032] Composite fabric **10** can be formed into any desired configuration. For example, as shown in FIG. 3, composite fabric **10** can be formed into a custom fit cover **100** that is specially configured to protect a vehicle **110** having a surface **111**. In use, retaining members **70** contact surface **111** of vehicle **110** (FIG. 4), thereby holding the remainder of cover **100** away from surface **111**. In this manner, should second fibrous layer **60** become wet, retaining members **70** hold it away from surface **111**, thereby keeping surface **111** drier and resisting corrosion. This configuration also creates pockets in which the VCI can collect. The pockets also assist with evaporation of condensation. Retaining members **70** also help secure cover **100** to vehicle **110**, such as when high winds push cover **100** toward or relative to vehicle **110** and the friction between retaining members **70** and surface **111** resists movement of cover **100**. In certain embodiments of the invention, retaining members **70** are capable of retaining cover **100** on vehicle **110** under wind speed conditions of from about 119 km/h to about 252 km/h. Furthermore, frictional heat generated by movement between retaining members **70** and surface **111** will facilitate the release of vapor corrosion inhibitors **71**, which helps prevent corrosion of vehicle **110** and surface **111**. Over time, some of the vapor corrosion inhibitors **51** will also migrate from fabric layer **50** toward surface **111** of vehicle **110** and help prevent corrosion.

[0033] FIG. 5 is a cross sectional view of a composite fabric **200** according to another embodiment of the present invention. Like the embodiment of FIGS. 1 and 2, this embodiment includes film **20**, adhesive **30**, first fibrous layer **40**, fabric layer **50**, second fibrous layer **60**, and retaining members **70**. However, composite fabric **200** also includes a second film **20A** between second fibrous layer **60** and retaining members **70**. Second film **20A** can be connected to second fibrous layer **60** by a second adhesive **30A** in the same manner as described above for connecting film **20** to first fibrous layer **40**. Second film **20A** can be constructed in the same manners and from the same materials discussed above in connection with film **20**. The pH modifiers **24** discussed above can also be utilized as pH modifiers **24A** in connection with second film **20A**. Note, however, that film **20** and film **20A** (as well as pH modifiers **24** and **24A**) of a particular fabric **200** do not have to be constructed in the same manner or from the same materials. Any of the adhesives discussed above that are suitable for use as adhesive **30** are also suitable for use as adhesive **30A**. However, different adhesives can be used as adhesive **30** and adhesive **30A** in any given composite fabric **200**. In another embodiment of the invention, retaining members **70** are attached to both film **20** and second film **20A**. Retaining members **70** may be printed onto film **20** and/or second film **20A** either before or after laminating the films to first fibrous layer **40** and second fibrous layer **60**.

[0034] Composite fabric **200** can be formed in any one of a number of manners. In one embodiment of the invention, first fibrous layer **40**, fabric layer **50**, and second fibrous layer **60** are first joined together, followed by laminating film **20** to first fibrous layer **40**. Retaining members **70** are then applied to second film **20A**, which is then laminated to second fibrous layer **60**. Alternatively, first fibrous layer **40**, fabric layer **50**, and second fibrous layer **60** are first joined together, followed by laminating film **20** to first fibrous layer **40** and laminating second film **20A** to second fibrous layer **60**. Retaining members **70** are then connected to second film **20A**.

[0035] In other embodiments of the invention, infrared (“IR”) shielding, radio frequency (“RF”) shielding, electromagnetic pulse (“EMP”) shielding, high-powered microwave (“HPM”) shielding, directed-energy weapons (“DEW”) shielding, and/or electromagnetic interference (“EMI”) shielding can be incorporated into fabrics and covers of the present invention. As used in this application, the terms “electromagnetic shielding” and “EMI shielding” include all of the foregoing types of shielding and all other types of shielding designed to block, inhibit, reduce or otherwise disrupt infrared, electronic, radio, microwave, static, magnetic and other similar forms of interference. Shielding electronics and other sensitive equipment from such interference helps

isolate electrical devices from the surroundings and may protect equipment from electronic attacks, such as RFID virus attacks in which a malicious code is inserted into an RFID tag to alter or corrupt data in an RFID system or equipment.

[0036] EMI shielding can be incorporated into composite fabric **10** in a number of ways. For example, conductive modifiers, such as carbon or metal, can be incorporated into one or more layers of composite fabric **10** to lower the surface and volume resistivity of the layers so that they can dissipate electrostatic charges and attenuate electromagnetic signals. The shielding materials can be incorporated into film, fibrous layers, fabric layer, and/or retaining members of embodiments of the present invention. The EMI shielding materials can be used to construct the film, fibrous layers, fabric layer, and/or retaining members of composite fabric **10**, can be incorporated as a component of those parts of composite fabric **10**, and/or can be applied to those parts of composite fabric **10** as a coating. In certain embodiments of the present invention, EMI shielding material is incorporated into adhesive **30**. In other embodiments, a separate layer, such as a fabric layer with EMI shielding functionality, is included in composite fabric **10** as an additional layer.

[0037] For example, in certain embodiments of the present invention, EMI shielding is provided by incorporating carbon modifiers into components of the present invention in amounts of 5 to 25%. Carbon modifiers in the form of fibers can be derived from polyacrylonitrile (“PAN”). In certain embodiments of the present invention, PAN carbon fibers are incorporated in amounts up to 10% by weight of composite fabric **10**. EMI shielding values of 30 dB to 50 dB can be achieved by incorporating approximately 50% by weight PAN carbon fibers into composite fabric **10**. Microcarbon fibers may achieve similar EMI shielding values at amounts as low as 3% by weight of composite fabric **10**. Other effective EMI shielding materials include nickel coated carbon fibers, stainless steel fibers (at levels of 5% to 10% by weight of composite fabric **10**), copper and aluminum.

[0038] In certain embodiments of the invention, EMI shielding effectiveness of between about 1 dB and about 100 dB or more can be achieved by utilizing different coatings that include aluminum, silver, nickel, copper or combinations of these materials. The coatings utilized can be solvent based or water based systems. One embodiment of the invention utilizes a water based silver coating, such as Series 599-Z6098-01 from PPG Industries, Inc. of Pittsburgh, Pennsylvania. The coating may be applied to the surface of the film, the surface of the fibrous layers, and/or the surface of the fabric layer. It can also be used to coat the individual strands of the fibrous and fabric layers. The coating system can also be added to the retaining members, used to coat the retaining members, and/or incorporated into the VCI system of the retaining members.

[0039] In certain embodiments of the present invention, one or more components of composite fabric **10** is constructed to form a Faraday cage or Faraday shield. This construction is particularly useful for blocking electromagnetic pulses, which can disrupt, degrade the performance of, and/or damage electronic components.

[0040] Covers made from composite fabrics **10** of the present invention can be manufactured by sewing, welding (such as, for example, impulse bar welding, ultrasonic welding, and RFID welding) and/or other methods. However, welded cover seams may result in better EMI shielding capabilities.

[0041] Although the present invention has been shown and described in detail, the same is by way of example only and should not be taken as a limitation on the invention. Numerous modifications can be made to the embodiments disclosed without departing from the scope of the present invention. For example, the protective cover can be formed into configurations other than those shown and described and can be used to protect objects other than vehicles. Also, vapor corrosion inhibitors could be incorporated into adhesive **30**. Other modifications and variations are also within the scope of the invention.

Claims

1. A composite fabric, comprising: a film having a first side and a second side; a first fibrous layer having a first side facing the second side of the film and a second side; a fabric layer having a first side facing the second side of the first fibrous layer and a second side; a second fibrous layer having a first side facing the second side of the fabric layer and a second side; and a plurality of retaining members extending from the second side of the second fibrous layer and configured to contact a surface on which the composite fabric is placed and to space at least a portion of the second side of the second fibrous layer from the surface, wherein the retaining members include a first vapor corrosion inhibitor within the plurality of retaining members configured to vaporize and collect between the surface and the portion of the second side of the second fibrous layer spaced from the surface wherein the plurality of retaining members and the second side of the second fibrous layer are configured to cooperate with the surface to form at least one pocket configured to retain the vaporized first vapor corrosion inhibitor therein.
2. The composite film of claim 1, wherein the plurality of retaining members are configured such that movement between the plurality of retaining members and the surface facilitates release of the first vapor corrosion inhibitor from the plurality of retaining members.
3. The composite film of claim 1, wherein the first fabric layer includes a second vapor corrosion inhibitor.
4. The composite film of claim 3, wherein the second vapor corrosion inhibitor is different than the first vapor corrosion inhibitor.
5. The composite film of claim 1, wherein the film comprises a non-heat shrinking film.
6. The composite fabric of claim 1, wherein the retaining members comprise polyurethane.
7. The composite fabric of claim 1, further comprising: a pH modifier.
8. The composite fabric of claim 1, further comprising: an adhesive positioned between and configured to connect the film and the first fibrous layer.
9. The composite fabric of claim 8, further comprising: a third vapor corrosion inhibitor in the adhesive.
10. The composite fabric of claim 1, wherein the film includes a multilayer film.
11. The composite fabric of claim 1, further comprising: an electromagnetic interference shielding layer positioned between the film layer and the retaining members.
12. The composite fabric of claim 1, wherein the plurality of retaining members is configured to resist movement of the composite fabric relative to the surface.
13. The composite fabric of claim 1, wherein the plurality of retaining members are configured to entirely space the second side of the second fibrous layer from the surface.
14. A composite fabric, comprising: a first film having a first side and a second side; a first fibrous layer having a first side facing the second side of the film and a second side; a fabric layer having a first side facing the second side of the first fibrous layer and a second side; a second fibrous layer having a first side facing the second side of the fabric layer and a second side; a second film having a second side facing the second side of the second fibrous layer and a second side; and a plurality of retaining members connected to and extending from the second side of the second film and configured to contact a surface on which the composite fabric is placed and space a portion of the second surface of the second film from the surface, wherein the retaining members include a corrosion inhibitor configured to collect between the second film and the surface, and wherein the plurality of retaining members form at least one pocket between one or more of the plurality of retaining members, the second side of the second film, and the surface, and wherein the first corrosion inhibitor vaporizes and is retained within one or more of the pockets.
15. The composite fabric of claim 14, wherein the corrosion inhibitor comprises a first vapor corrosion inhibitor.

- 16.** The composite fabric of claim 14, further comprising: a pH modifier.
 - 17.** The composite fabric of claim 14, further comprising: an adhesive between and connecting the film and the first fibrous layer, wherein the adhesive includes a second vapor corrosion inhibitor.
 - 18.** The composite fabric of claim 14, further comprising: an electromagnetic interference shielding layer positioned between the film layer and the retaining members.
 - 19.** The composite fabric of claim 14, wherein the corrosion inhibitor is one of a plurality of corrosion inhibitors within the fabric, and wherein the plurality of corrosion inhibitors differ in chemical composition from one another.
 - 20.** The composite fabric of claim 14, wherein the first and second films comprise a non-heat shrinking film.
 - 21.** A composite fabric, comprising: a first film having a first side and a second side; a fabric layer having a first side and a second side; at least one fibrous layer; a second film having a first side facing the first film layer and a second side, wherein the fabric layer and the at least one fibrous layer is positioned between the first and second film layers; and a plurality of retaining members configured to contact a surface on which the composite fabric is placed, the retaining members connected to and extending from the second side of the second film, wherein the retaining members include a first vapor corrosion inhibitor, and wherein the plurality of retaining members form at least one pocket between one or more of the plurality of retaining members, the second side of the second film, and the surface, wherein at least a portion of the first vapor corrosion inhibitor collects in one or more of the pockets.
 - 22.** The composite fabric of claim 21, wherein the at least one fibrous layer includes a first fibrous layer having a first side facing the second side of the first film and a second side facing the fabric layer and a second fibrous layer that includes a first side facing the fabric layer and a second side facing the first side of the second film.
 - 23.** The composite fabric of claim 22, further comprising: an adhesive connecting the first and second films to the first and second fibrous layers, respectively.
 - 24.** The composite fabric of claim 23, further comprising: a second vapor corrosion inhibitor in the adhesive.
 - 25.** The composite fabric of claim 21, wherein the plurality of retaining members are configured such that movement between the plurality of retaining members and the surface facilitates release of the first vapor corrosion inhibitor from the plurality of retaining members.
 - 26.** The composite fabric of claim 21, wherein the first and second film each comprise a non-heat shrinking film.
 - 27.** The composite fabric of claim 21, wherein the retaining members comprise polyurethane.
 - 28.** The composite fabric of claim 21, further comprising: a pH modifier.
 - 29.** The composite fabric of claim 21, further comprising: electromagnetic shielding.
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